$$N = \left(\frac{1}{2}\right)^n N_0 = \left(\frac{1}{2}\right)^4 \times 10,000$$

$$N = 625$$

EXAMPLE 9.13

A radioactive sample has $2.6\mu g$ of pure $^{13}_{7}N$ which has a half-life of 10 minutes. (a) How many nuclei are present initially? (b) What is the activity initially? (c) What is the activity after 2 hours? (d) Calculate mean life of this sample.

Solution

(a) To find N_0, we have to find the number of $^{13}_{7}N$ atoms in $2.6\mu g$. The atomic mass of nitrogen is 13. Therefore, 13 g of $^{13}_{7}N$ contains Avogadro number (6.02×10^{23}) of atoms.

In 1 g, the number of $^{13}_{7}N$ atoms present is equal to $\frac{6.02\times10^{23}}{13}$. So the number of $^{13}_{7}N$ atoms present in $2.6\mu g$ is

$$N_0 = \frac{6.02\times10^{23}}{13}\times2.6\times10^{-6} = 12.04\times10^{16}$$

(b) To find the initial activity R_0, we have to evaluate decay constant λ

$$\lambda = \frac{0.6931}{T_{1/2}} = \frac{0.6931}{10\times60} = 1.155\times10^{-3}\,s^{-1}$$

Therefore

$$R_0 = \lambda N_0 = 1.155\times10^{-3}\times12.04\times10^{16}$$
$$= 13.90\times10^{13}\,\text{decays/s}$$
$$= 13.90\times10^{13}\,\text{Bq}$$

In terms of a curie,

$$R_0 = \frac{13.90\times10^{13}}{3.7\times10^{10}} = 3.75\times10^{3}\,\text{Ci}$$

since $1\text{Ci} = 3.7\times10^{10}\text{Bq}$

(c) Activity after 2 hours can be calculated in two different ways:

Method 1: $R = R_0 e^{-\lambda t}$

At $t = 2\,\text{hr} = 7200\,\text{s}$

$$R = 3.75\times10^3\times e^{-7200\times1.155\times10^{-3}}$$

$$R = 3.75\times10^3\times2.4\times10^{-4} = 0.9\,\text{Ci}$$

Method 2: $R = \left(\frac{1}{2}\right)^n R_0$

Here $n = \frac{120\,\text{min}}{10\,\text{min}} = 12$

$$R = \left(\frac{1}{2}\right)^{12}\times3.75\times10^3 \approx 0.9\,\text{Ci}$$

(d) mean life $\tau = \frac{T_{1/2}}{0.6931} = \frac{10\times60}{0.6931}$
$= 865.67\,s$

9.6.6 Carbon dating

The interesting application of beta decay is radioactive dating or carbon dating. Using this technique, the age of an ancient object can be calculated. All living organisms absorb carbon dioxide (CO_2) from air to synthesize organic molecules. In this absorbed CO_2, the major part contains $^{12}_{6}C$ and very small fraction (1.3×10^{-12}) contains radioactive $^{14}_{6}C$ whose half-life is 5730 years.

Carbon-14 in the atmosphere is always decaying but at the same time, cosmic rays from outer space are continuously bombarding the atoms in the atmosphere which produces $^{14}_{6}C$. So the continuous production and decay of $^{14}_{6}C$ in the atmosphere keep the ratio of $^{14}_{6}C$ to $^{12}_{6}C$ always constant. Since our human body, tree or any living organism continuously absorb CO_2 from the atmosphere, the ratio of $^{14}_{6}C$ to $^{12}_{6}C$ in the living organism is also nearly constant. But when the organism

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